

Quiz 2

1. Answer briefly as specified:

- (a) TRUE or FALSE (strike off):
 - i. Backpropagation, properly implemented, will always yield global minima.
 - ii. Backpropagation works well when weights are initialized as zeros.
 - iii. Cross entropy is popular for regression problems.
- (b) Write precise answer (i.e., equations and notations):
 - i. Write the simple gradient descent update equation for obtaining w^{k+1} from w^k . Define any new symbols.
 - ii. Write the gradient descent with momentum update equation for obtaining w^{k+1} from w^k . Define any new symbols.
 - iii. Write the update equation for obtaining w^{k+1} from w^k using Newton's method. Define any new symbols.
- (c) A dataset D consists of six feature measurements:

$$D = \{43, 45, 49, 51, 56, 57\}$$

D needs to be normalized such that mean is zero and variance is unity. Write the normalized D .

- (d) Test your breadth:
 - A famous AI researcher who recently left Meta and founded his own start-up and who has made pioneering contributions in CNNs: -----
 - Expand the popular acronym ReLU.
 - Which company is the primary manufacturer of GPUs used in training deep AI models?
- (e) Consider an MLP with two inputs, one output and one hidden layer with eight neurons. Assuming no bias, how many learnable parameters are there in this MLP?

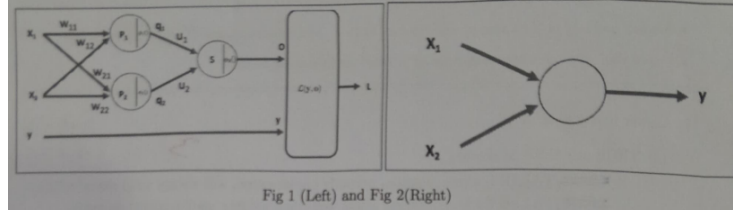


Figure 1: Fig 1 (Left) and Fig 2 (Right)

2. Consider Fig 1. We have an MLP with MSE Loss:

$$\mathcal{L}(y, o) = \frac{1}{2}(y - o)^2$$

$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}, \quad u = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

and also $\phi_1(\cdot)$ (in the hidden layer) as ReLU and $\phi_2(\cdot)$ (in the output) linear.

- Find o if $x = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$.
 - Write $\frac{\partial \mathcal{L}}{\partial w_{11}}$ as a product of multiple terms (i.e., using chain rule).
 - Numerically compute $\frac{\partial \mathcal{L}}{\partial w_{11}}$. Assume $y = 3$ for $x = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$.
 - Let us revisit the MLP in Fig 1. It is often argued that if $\phi_1(\cdot)$ is linear, we can represent the MLP as an SLP. Draw SLP and mark weights (numerically) as in Fig 2.
3. Consider the problem of linear classification with a single layer perceptron (Fig 2) with three training examples:

$$D = \left\{ \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, +1 \right), \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, +1 \right), \left(\begin{bmatrix} -1 \\ -1 \end{bmatrix}, -1 \right) \right\}$$

The weights for these two dimensions are initialized as $w_1 = -0.8$, $w_2 = -0.8$, and the bias is initialized as $w_0 = 0$.

- What is the classification rule of this SLP (clearly specify the activation function)? What is the output of the classifier on these three training samples and what is the error on this set?
- Write the perceptron training algorithm with appropriate expressions/notations. (Write as simple steps)
- Use perceptron algorithm and update the weights. Show two iterations with a learning rate of 0.5.